

Shu Zhang

Contact & Personal Information

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Research Profile

Dr. Zhang is Assistant Professor of the Institute of Energy, Environment and Economy at Tsinghua University. His research focuses on integration and interactions across energy, land, water, and air quality systems, with particular emphasis on cross-sector synergies. He develops and applies the energy-environment-economy model China TIMES to explore pathways that integrate energy, transport, buildings, and industry sectors, while accounting for shared infrastructures such as power grid, storage and CO₂ transport network. A central strand of his research is quantifying uncertainty in decarbonization pathways using the probabilistic framework, which enables robust policy recommendations. With publications in Nature Sustainability, Nature Communications, and Engineering, He has contributed new insights on the costs, potentials, and enabling policies of integrated low-carbon transitions, and provides evidence-based insights to support strategic policymaking.

Professional Experiences

Jul 2024 – present 📌 **Assistant Professor**
Institute of Energy, Environment and Economy, Tsinghua University, China

Mar 2022 – Apr 2023 📌 **Visiting Scholar**
International Institute for Applied Systems Analysis (IIASA), Austria

Education

Aug 2019 – Jun 2024 📌 **Ph.D., Tsinghua University, China** in Management Science & Engineering
Thesis title: Assessing the Energy Transition, Synergies and Uncertainties toward Carbon Neutrality in China (Supervisor: Prof. Wenyong Chen)

Aug 2015 – Jul 2019 📌 **B.Eng., Tsinghua University, China** in Electrical Engineering (Major)
Core field: Power load forecasting (Supervisor: Prof. Chongqing Kang)

B.Econ., Tsinghua University, China in Economics (2nd degree)
Core field: Behavioral Economics

B.Fin., Tsinghua University, China in Finance (Minor)

Grants & Research Funding

As PI or co-PI

Mar 2026 – Sep 2026 📌 **International (Regional) Cooperation and Exchange program, National Natural Science Foundation of China (NSFC)**
Special Seminar on Climate Change Assessment in Hong Kong and Macao, Principal Investigator

- Jan 2026 – Dec 2028  **Young Scientists Fund (C), National Natural Science Foundation of China (NSFC)**
Hydrogen Energy and Hydrogen-based Fuel Supply Potential and Development Pathway in China under the Carbon Neutrality Target, Principal Investigator
- Sep 2025 – Sep 2026  **International Joint Mission on Climate Change and Carbon Neutrality**
China's green hydrogen, green ammonia and green methanol fuel development pathways under the carbon neutrality target, Principal Investigator

As Member

- May 2026 – Dec 2028  **China National Nuclear Corporation**
Methodological research on voluntary greenhouse gas reduction projects in the nuclear energy industry, Key Member
- Mar 2026 – Feb 2029  **National Key Research and Development Program, Ministry of Science and Technology of China (MOST)**
New pathways for China's carbon neutrality considering equitable climate responsibility and sustainable development, Key Member
- Jan 2025 – Jun 2028  **Horizon Europe, European Commission**
New Pathways for Equitable Climate Action in Line with the Paris Agreement and Sustainable Development (NEWPATHWAYS), Key Member
- Nov 2024 – Dec 2030  **National Science and Technology Major Project, Ministry of Science and Technology of China (MOST)**
CO₂ driven technology for significant enhanced oil recovery and long-term sequestration, WP6: Planning Study for CO₂ Oil Diversion and Sequestration Development, Key Member
- Aug 2024 – May 2025  **China National Nuclear Corporation**
Methodological research on carbon emission accounting for the nuclear energy industry chain, Key Member
- Mar 2023 – Jun 2026  **DG-CLIMA Project, European Commission**
Enhanced sharing of good practices on greenhouse gas emissions modelling between EU and Asian countries (COMMITTED), Key Member
- Jan 2023 – Oct 2025  **Key Research Base for Humanities and Social Sciences Major Project, Ministry of Education (MOE)**
Research on Electricity Transition Pathways and Policies under the Carbon Neutral Vision 2060, Key Member
- Jun 2021 – Jun 2023  **World Bank Group Project**
Green China: Towards Cleaner and More Sustainable Growth, Key Member
- Mar 2019 – Feb 2022  **International Cooperation and Exchange Project, National Natural Science Foundation of China (NSFC)**
Socio-Techno-Economic Pathways for sustainable Urban energy development (STEP-UP), Key Member
- Jan 2019 – Dec 2023  **Horizon 2020, European Commission**
Exploring National and Global Actions to reduce Greenhouse gas Emissions (ENGAGE), Key Member

- Jan 2019 – Dec 2021 ■ **National Key Research and Development Program, Ministry of Science and Technology of China (MOST)**
Research on Global Governance and Domestic Responses to Climate Change Risks, WP6: Simulation of synergistic governance pathways to address climate change and the economy, society and environment in China, Key Member
- Jan 2018 – Dec 2021 ■ **International Cooperation and Exchange Project, National Natural Science Foundation of China (NSFC)**
Multi-model innovations in Integrated Assessment Modelling of Global, Chinese, and Irish energy-economy-environment-climate systems investigating deep decarbonization pathways from the Paris Agreement to the United Nations sustainable development goals (CHIMERA), Key Member
- Jan 2017 – Dec 2021 ■ **Major Program, National Natural Science Foundation of China (NSFC)**
Research on Green and Low-carbon Economy Transformation Management and Policy, WP4: Study on international climate governance and cooperation mechanisms, Key Member
- Sep 2015 – Sep 2019 ■ **Horizon 2020, European Commission**
Linking Climate and Development Policies-Leveraging International Networks and Knowledge Sharing (CD-LINKS), Key Member

Research Publications

Journal Articles

- 1 **Zhang, S.**, Chen, W.* (2026). Multi-model intercomparison study of China's low carbon transition pathways under the "Dual Carbon" goal. *Climate Change Research*, <https://doi.org/10.12006/j.issn.1673-1719.2026.133>. (In Chinese)
- 2 **Zhang, S.**, and Chen, W.* (2026). Probabilistic scenarios reveal the impacts of China's energy system net-zero transition on the water-energy-food nexus. *Environmental Science & Technology* 60, 22377–22388, <https://doi.org/10.1021/acs.est.6c03030>.
- 3 Ding, Q., Ren, J., **Zhang, S.**, and Chen, W.* (2026). The Role of Shared Autonomous Electric Vehicles in Decarbonizing China's Passenger Transport Sector. *Applied Energy* 422, 128259, <https://doi.org/10.1016/j.apenergy.2026.128259>.
- 4 Lyu, J., Ren, J., **Zhang, S.**, and Chen, W.* (2026). Building a net-zero power system under future climate conditions with data-driven forecasting and adequacy assessment. *Energy* 356, 141119, <https://doi.org/10.1016/j.energy.2026.141119>.
- 5 **Zhang, S.**, and Chen, W.* (2026). Exploring the feasible net-zero transition pathway in China considering energy system flexibility. *Nature Communications* 17, 5440, <https://doi.org/10.1038/s41467-026-71410-2>.
- 6 Lyu, J., **Zhang, S.**, Chen, W.* (2026). The Development Process of CCUS in China Towards Carbon Neutrality: Insights from International Experience. *Climate Change Research* 22, 70–83, <https://doi.org/10.12006/j.issn.1673-1719.2025.115>. (In Chinese, **CNKI High Impact Paper**)
- 7 Dong, H.*, Zhang, T., Geng, Y., Wang, P., **Zhang, S.**, and Sarkis, J. (2025). Sub-technology market share strongly affects critical material constraints in power system transitions. *Nature Communications* 16, 1285, <https://doi.org/10.1038/s41467-025-56592-5>.

- 8 Chen, W.*, **Zhang, S.**, Zhang, Q., Ren, J., and Ding, Q. (2025). Assessing China's province-coordinated power system carbon-neutral transition pathway. *Journal of Energy and Climate Change* 1, 1–15, <https://doi.org/10.3724/j.issn.2097-4981.JECC-2024-0022>. (In Chinese) (**Best Paper in 2025**)
- 9 Zhang, Q., **Zhang, S.**, and Chen, W.* (2024). Provincial pathways to carbon-neutral energy systems in China considering interprovincial electricity transmission development. *Applied Energy* 375, 123953, <https://doi.org/10.1016/j.apenergy.2024.123953>.
- 10 **Zhang, S.**, Chen, W.*, Zhang, Q., Krey, V.*, Byers, E., Rafaj, P., Nguyen, B., Awais, M., and Riahi, K. (2024). Targeting net-zero emissions while advancing other sustainable development goals in China. *Nature Sustainability* 7, 1107–1119, <https://doi.org/10.1038/s41893-024-01400-z>. (**ESI Highly Cited Paper, ESI Hot Paper, Featured paper**)
- 11 Tang, H., Chen, W.*, **Zhang, S.**, and Zhang, Q. (2023). China's multi-sector-shared CCUS networks in a carbon-neutral vision. *iScience* 26, 106347, <https://doi.org/10.1016/j.isci.2023.106347>.
- 12 Shao, T., Pan, X.*, Li, X., Zhou, S., **Zhang, S.**, and Chen, W. (2022). China's industrial decarbonization in the context of carbon neutrality: A sub-sectoral analysis based on integrated modelling. *Renewable and Sustainable Energy Reviews* 170, 112992, <https://doi.org/10.1016/j.rser.2022.112992>.
- 13 **Zhang, S.**, and Chen, W.* (2022). China's Energy Transition Pathway in a Carbon Neutral Vision. *Engineering* 14, 64–76, <https://doi.org/10.1016/j.eng.2021.09.004>. (**ESI Highly Cited Paper, ESI Hot Paper, Best Paper in 2022**)
- 14 **Zhang, S.**, and Chen, W.* (2022). Assessing the energy transition in China towards carbon neutrality with a probabilistic framework. *Nature Communications* 13, 87, <https://doi.org/10.1038/s41467-021-27671-0>. (**ESI Highly Cited Paper, ESI Hot Paper, Top 25 Environment Paper**)
- 15 Tang, H., **Zhang, S.**, and Chen, W.* (2021). Assessing Representative CCUS Layouts for China's Power Sector toward Carbon Neutrality. *Environmental Science & Technology*. 55, 11225–11235, <https://doi.org/10.1021/acs.est.1c03401>.
- 16 **Zhang, S.**, Wang, Y.*, Zhang, Y., Wang, D., and Zhang, N. (2020). Load probability density forecasting by transforming and combining quantile forecasts. *Applied Energy* 277, 115600, <https://doi.org/10.1016/j.apenergy.2020.115600>.

Conference Proceedings

- 1 **Zhang, S.**, and Chen, W.* (2020). Modeling the Rapid Development of Electric Vehicles and Energy Storage Technology Under China Carbon Neutral Scenario Based on China-TIMES Model. In J. Yan, ed. *Proceedings of 12th International Conference on Applied Energy*. <https://doi.org/10.46855/energy-proceedings-7202>

Working Papers

- 1 **Zhang, S.**, Zhang, Q., Chen, W.* (2026). China's Hydrogen-Based Energy Development and Policy Requirements Toward Carbon Neutrality. *Climate Change Research*. (In Chinese)
- 2 Zhang, Q.†, **Zhang, S.**†, Gong, C., and Chen, W.* (2026). Hourly matching identifies hidden flexibility costs in renewable hydrogen-derived fuel production. *Science Bulletin*. (In Peer Review)
- 3 Vishwanathan, S. *, Garg, A., et al. (2026). Commonalities and differences in national pathways toward net-zero emissions, *Nature Climate Change*. (In Peer Review)
- 4 Vishwanathan, S. *, Garg, A., et al. (2026). National Scenarios Dataset 2025 - Business As Usual, Current Policies and Net-Zero, *Scientific Data*. (In Peer Review)

- 5 Shao, T.†, Wang, J.†, Pan, X.†*, **Zhang, S.**, et al. (2026). LLM Agent-informed 1.5 °C Global Mitigation Pathways Considering Sustainable Development Goals, *Nature Climate Change*. (In Peer Review)
- 6 Deng, N., Wang, Z., et al. (2026). Hidden Emissions from Coal-Based Grid Flexibility under Renewable Integration, *Nature Communications*. (In Peer Review)
- 7 Lyu, J., Chen, G., **Zhang, S.**, and Chen, W.* (2026). Hourly consumption-based carbon intensity reveals spatiotemporal patterns and load-carbon mismatch in China's high-renewable power system. *Applied Energy*. (In Peer Review)
- 8 Ren, J., Zhang, Q., Zhang, S., **Zhang, S.**, and Chen, W.* (2026). Climate-driven impacts in renewable resources alter green hydrogen supply. *Nature Climate Change*. (In Peer Review)

Professional Service

Editorial roles Energy and Climate Management – Editor

Journal Reviewer Nature Sustainability, Nature Climate Change, Renewable & Sustainable Energy Reviews, Applied Energy, Geoscientific Model Development, Progress in Energy, Communications Earth & Environment, npj Climate Action, Climatic Change, Journal of Cleaner Production, International Journal of Environmental Science and Development

Scientific Report Fifth National Assessment Report on Climate Change: Climate Change Mitigation (Chap. 1: Assessment framework and methodology, Chap. 2: China's transition pathway toward carbon neutrality), Ministry of Science and Technology of China (MOST), Lead Author

Fifth National Assessment Report on Climate Change: Special Report on Greenhouse Gas Scenario Framework (Chap. 4: Differences in scenarios and key influencing factors), Ministry of Science and Technology of China (MOST), Lead Author

Seventh Assessment Report (AR7) Climate Change: Mitigation of Climate Change (Chap. 3: Projected futures in the context of sustainable development and climate change), The Intergovernmental Panel on Climate Change (IPCC), Contributing Author

Synthesis Report 2022 on China's Carbon Neutrality: Electrification in China's Carbon Neutrality Pathways, Energy Foundation China, Lead Author